

LUKE MCBEE

MECHANICAL ENGINEERING CANDIDATE

Contact

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Columbia, MO 65203 

Professional Summary

I am a dedicated mechanical engineering student seeking an entry level position that will help me apply my skills and knowledge in a hands-on environment where I can make an immediate impact. I demonstrate proficiency in CAD software including SolidWorks and Autodesk Inventor, as well as Microsoft Office programs. I also possess a strong interest in problem-solving, mechanical design, and have a major focus on precision and quality in all my work.

Skills

- **CAD:** SolidWorks - Autodesk Inventor - Autodesk Revit
- **Engineering Analysis:** Fluid Flow Analysis - Vehicular Dynamic Analysis - Materials and Stress Analysis
- **Engineering Design:** Mechanical Design - Automotive Systems Design - Circuit Design
- **Software:** MATLAB - Microsoft Excel - Microsoft Word
- **Professional:** Team Leadership - Problem Solving – Communication - Project Management

Education

BACHELOR OF SCIENCE (B.S.) IN MECHANICAL ENGINEERING CANDIDATE

GREENVILLE UNIVERSITY — Greenville, IL, USA

- Expected graduation May 2027
- Engineering Honors Scholarship
- GPA 3.66

Notable Courses

- **Vehicle Dynamics** - Learned in depth engineering dynamics concepts as they relate to automobiles and learned how to analyze automotive performance and how specific components function.
- **Engineering Design and CAD** - Developed proficient skills in computer aided design software and learned how to use SolidWorks drafting software for single parts, mechanical assemblies, and analysis simulations.
- **Thermodynamics and Fluid Dynamics** - Learned in depth mechanical engineering principles and gained experience in analyzing and modeling different kinds of real-world systems using SolidWorks simulations for fluid flow.

Engineering Projects

Automotive Lift Weight Shift Sensor

- Designing and developing a product for a senior design capstone course.
- Working in a role as design and documentation/report lead.
- Working alongside two other engineers to design, develop, market, and patent this product.

Vehicle Aerodynamics and Fluid Flow Analysis

- Developed a SolidWorks model of a simplified race car and performed external-flow simulation.
- Modified vehicle geometry to increase theoretical aerodynamic downforce.
- Compared analytical calculations with simulation results and identified any errors in the theoretical calculations.

Automotive Component Reverse Engineering – Chevrolet SSR Fog Light Bezel

- Measured an existing automotive component and developed a full-scale 3D CAD model in SolidWorks.
- Applied reverse-engineering techniques to reproduce a discontinued replacement component.
- Created a digital model suitable for future manufacturing through 3D printing or machining.

Work Experience

CONSTRUCTION ASSISTANT AND PROPERTY CARE

May 2026 – Aug 2026

HEMME CONSTRUCTION — Columbia, MO

- Maintained and troubleshoot small engines, power equipment, and construction tools.
- Performed hands-on repair and maintenance to keep equipment operational.
- Assisted with construction and finishing operations while meeting project deadlines.
- Collaborated with multiple departments to complete construction projects efficiently.

HEAD BASEBALL COACH

May 2025 - Present

FIVE TOOL PERFORMANCE BASEBALL — Columbia, MO

- Mentor and develop players while establishing individualized improvement strategies.
- Communicate with players and parents regarding performance, development, and team expectations.
- Lead practices and training while managing multiple responsibilities and deadlines.

Other Activities

- **NCAA Division III Baseball Player:** 2023-Present